

Design of IOT Architecture for Battery Charging Management System

Battery Swapping station and EV Charging Infrastructure

Engineering & Technology



Abstract

Rapid growing of electric vehicle (EV) adoption faces challenges such as range anxiety and inconsistent charging infrastructure, which can be effectively addressed through an IoT-based architecture that offers real-time monitoring, predictive analytics, and mobile charging units, ultimately enhancing user experience, optimizing energy management, and promoting sustainable energy solutions for a cleaner future.

Introduction

•Growing Demand for Charging Infrastructure:

The rapid increase in electric vehicle (EV) sales is creating a pressing need for more efficient and accessible charging stations to alleviate range anxiety among users.

•Integration of Renewable Energy:

There is a critical requirement for sustainable charging solutions that incorporate renewable energy sources, such as solar power, to align with environmental sustainability goals.

•Cost Reduction and Innovation:

The necessity to reduce operational costs in charging management is driving the adoption of innovative IoT technologies aimed at optimizing battery management and enhancing overall efficiency.

Problem Statement

- Demand of Charging Infrastructure Expansion
- Grid Load Disturbances
- High Installation and Operational Costs
- Inefficient Load Management
- Limited Remote Monitoring

Objectives

- Develop an IoT-based architecture for optimized battery charging management.
- Incorporate real-time data monitoring and predictive analytics.
- Enhance operational cost efficiency by up to 88.5%.
- Promote sustainable energy integration through solar and ESS.

Motivation

The rapid growth of electric vehicles (EVs) necessitates a robust and efficient charging infrastructure that integrates renewable energy sources like solar power for environmental sustainability while driving innovation in IoT-based battery management technologies to reduce operational costs.

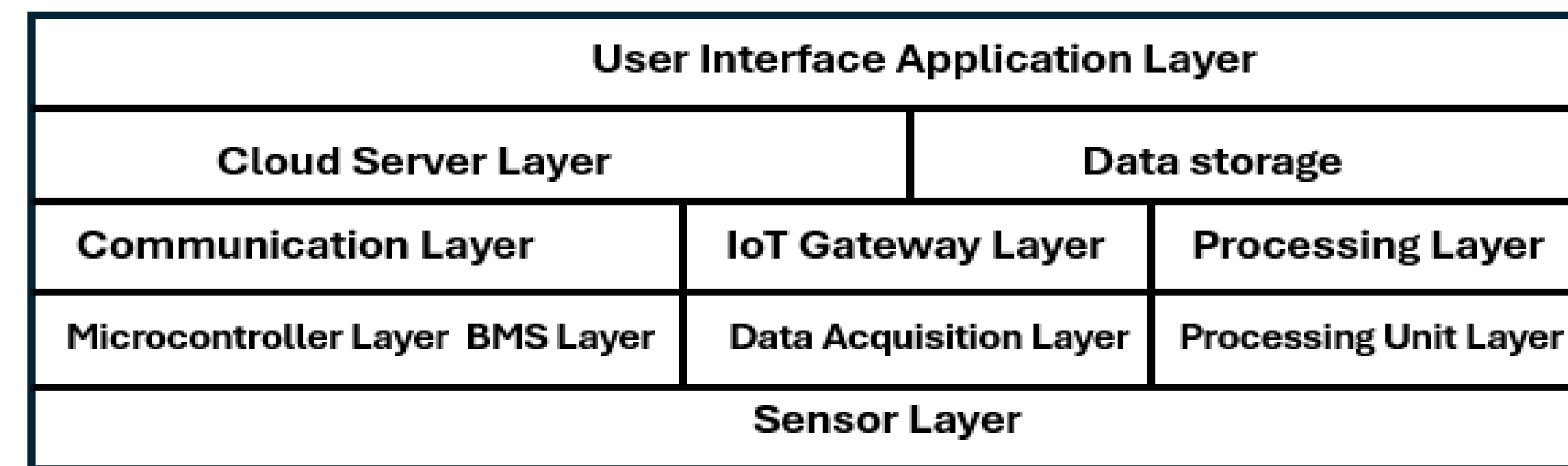
Project Novelty

- Innovative IOT 5 layered Architecture.
- Integration of multiple energy sources.
- Central monitoring and controlling.
- Real-Time data processing and user interface.
- Cloud connectivity.
- Cost efficiency model.
- Scalability and Adaptability.
- Portable charging station.

Methodology

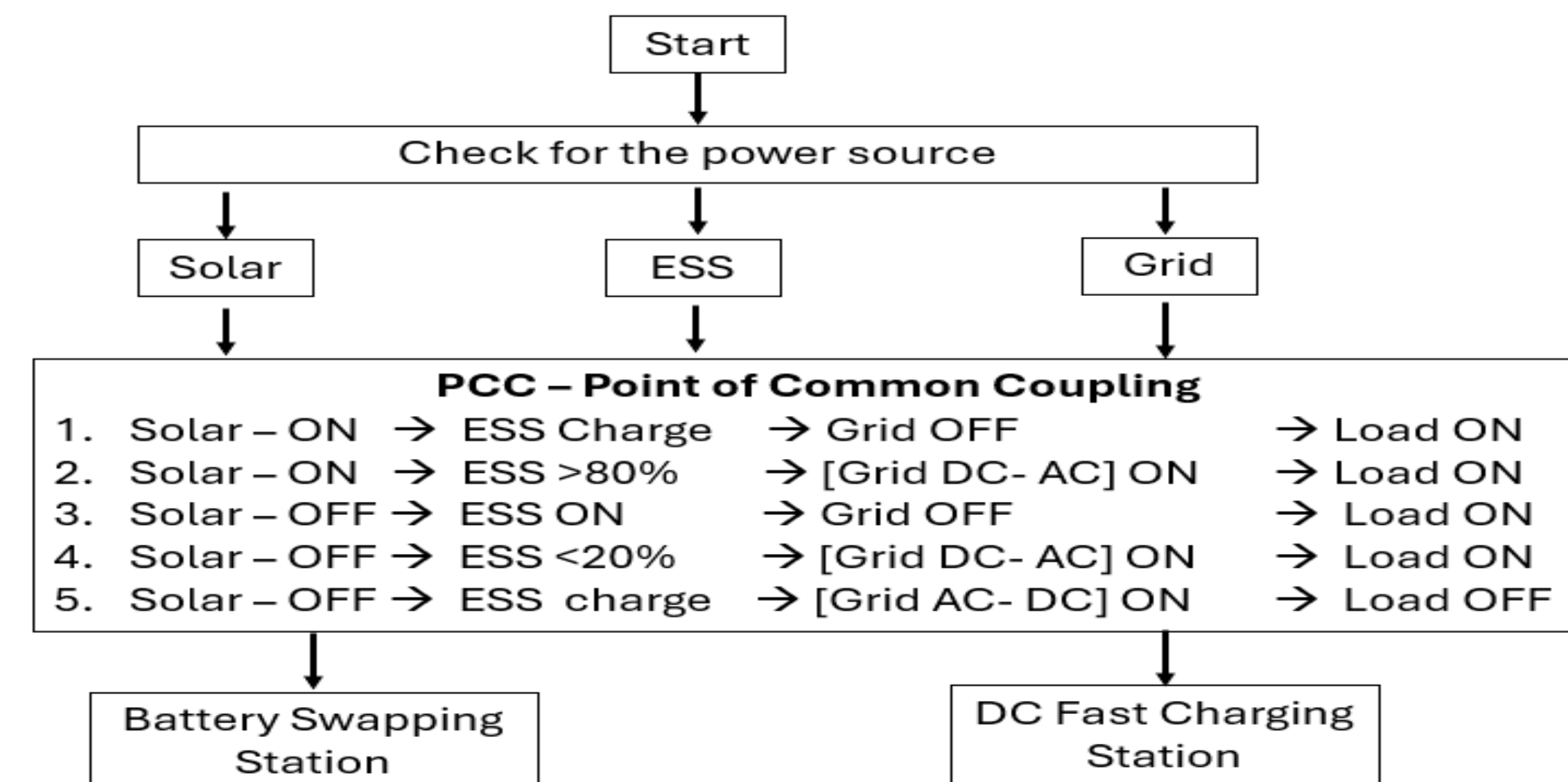
IoT Five-Layered Architecture

- User Interface Application Layer
- Cloud Server and Data Storage
- Communication Layer
- Microcontroller Layer
- Sensor Layer.



Point of Common Coupling (PCC)

Combines solar power, ESS, and grid power for Integration and Automation.



Algorithms

MPPT (Maximum Power Point Tracking) Algorithm:

Optimizes solar panel output by continuously adjusting the operating point to maximize energy capture under varying conditions.

Battery Swapping Station Algorithm:

Manages logistics and scheduling for battery swaps, optimizing inventory and availability of charged batteries to improve service efficiency.

Vehicle Charging Control Calculation:

Calculates optimal charging rates based on battery state-of-charge (SOC) and user demand, ensuring safe and efficient charging.

Safety and Blast Condition Prediction using ML:

Utilizes machine learning models to analyze historical data and predict potential safety hazards in battery charging environments, enabling proactive risk management.

Cloud Integration AWS

Secured MQTT Protocol:

- Utilizes MQTT for secure and
- Efficient communication between IoT devices and the cloud.

Data Storage:

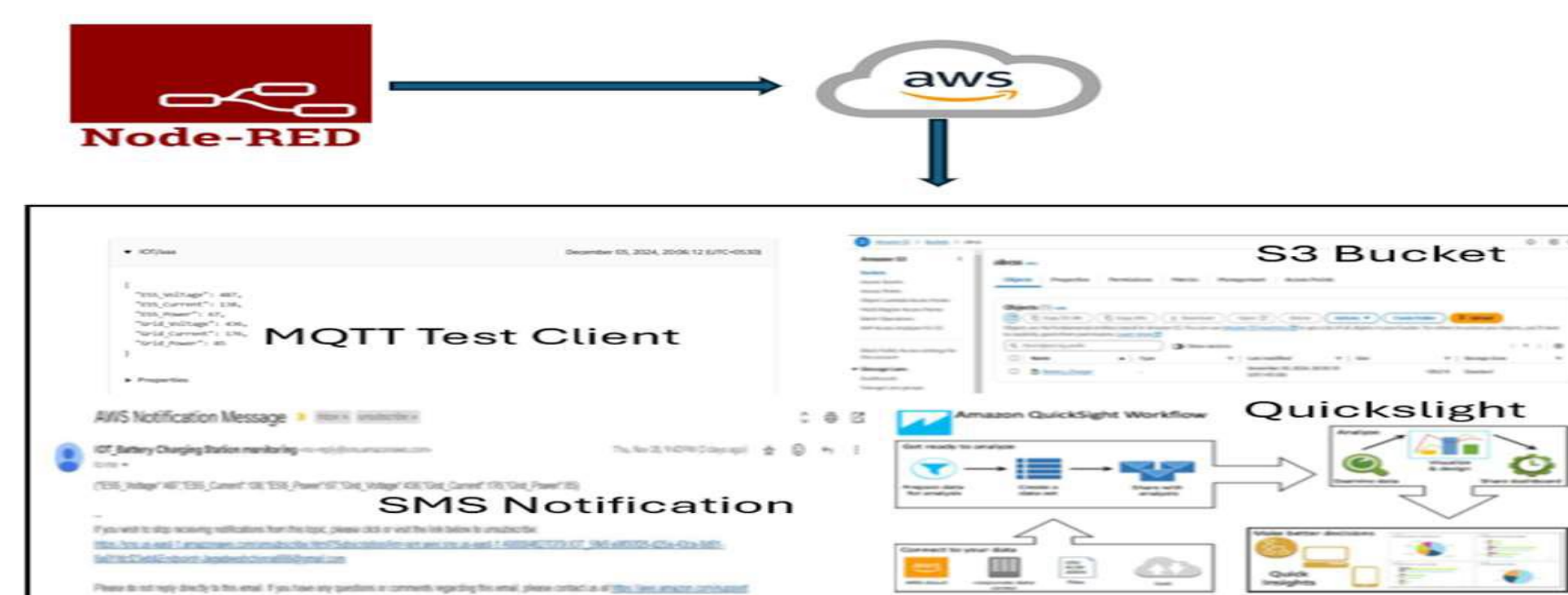
- S3 Bucket and durable storage solutions.
- Stores collected data in Amazon S3 for scalable.

Email Alerts:

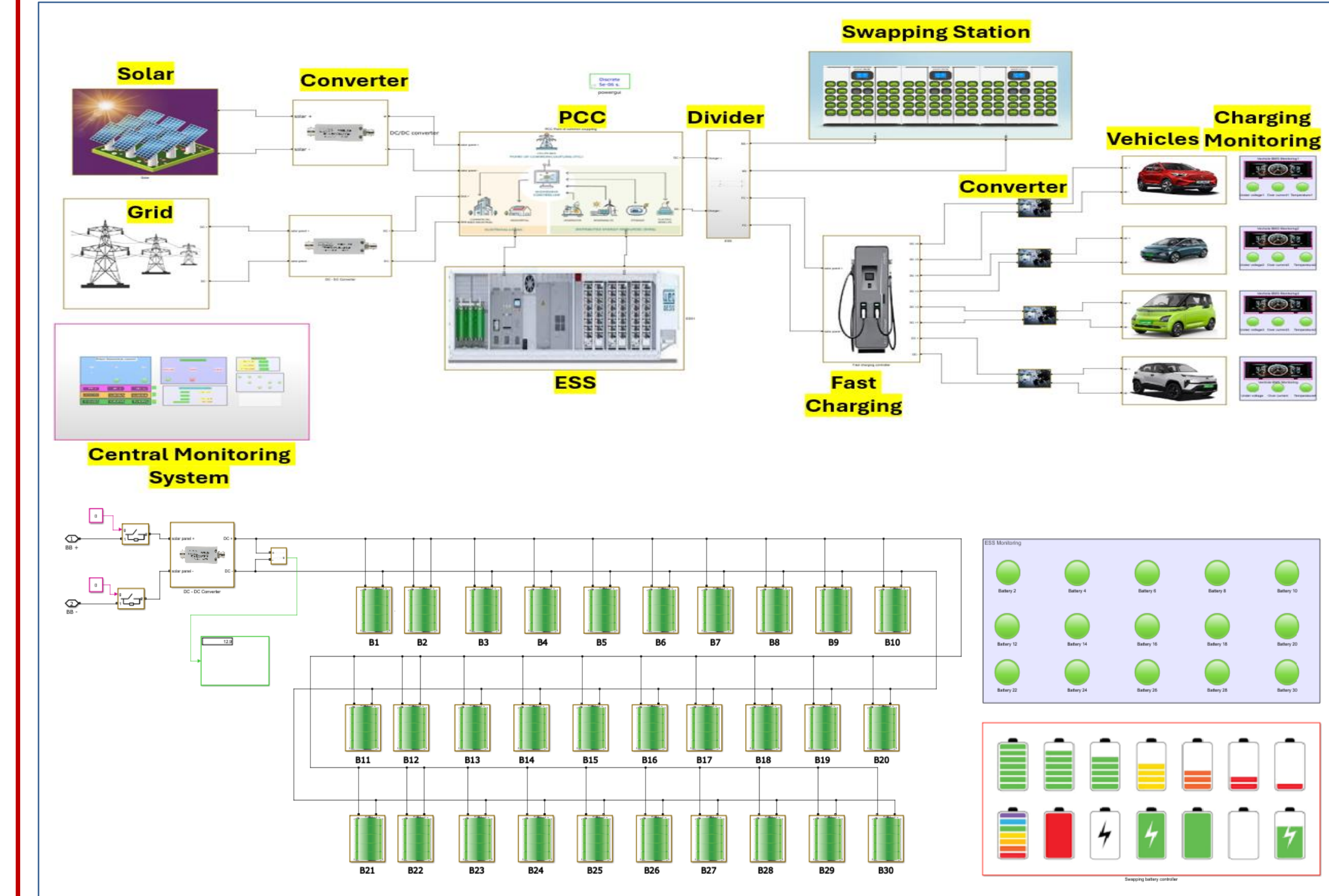
- Implements automated email notifications for critical events or alerts related to battery charging management.

Data Visualization:

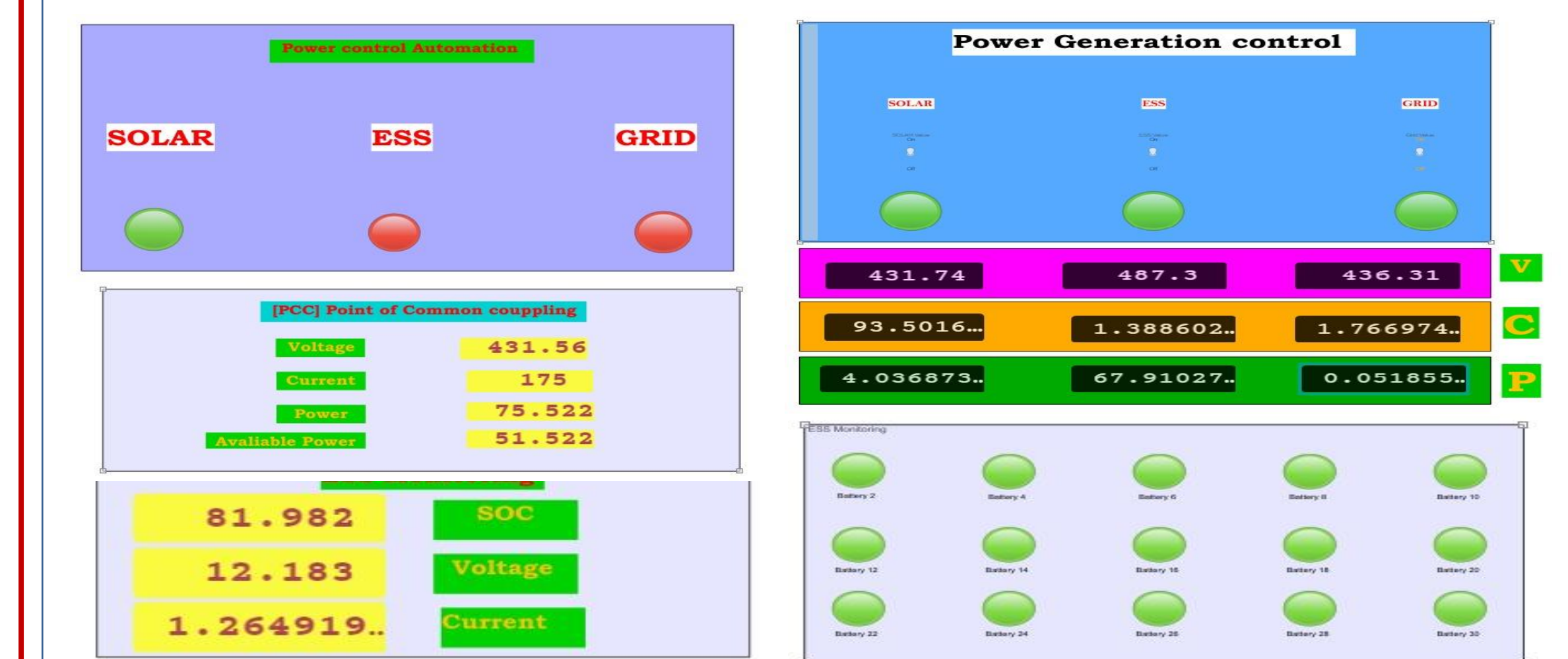
- Provides real-time data visualization dashboards to monitor charging status and performance metrics.



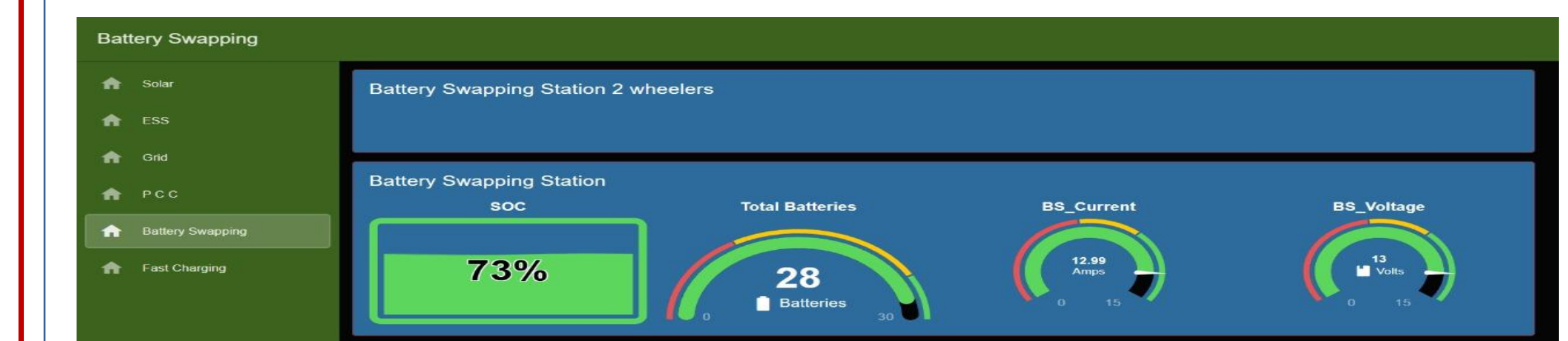
Simulation work



Real Time Monitoring



Data Visualization



Estimated Energy and Cost calculation

Power Generation						
Power Source	Voltage [V]	Current [I]	% of Generation/day	Required Power	Power [P]- KW/hr	Power/day
Solar	430	16.7	60	72	43	1032
ESS	480	16.7	30	80	24	576
Grid	435	20.0	10	87	9	209
Total Power Generation per day in KW/24 hr						1817
PCC - Point of common coupling			Battery Swapping Station			
PCC will Logically controlled on availability of source			Total Batteries 30			
			Under maintenance 0			
			Available batteries 30			
			Voltage 12			
			Current 10			
			Time / Battery in hr 5			
			No of Cycle per day 3			
			Power/Battery - KW 0.12			
			Total Station KW 18			
			Total Energy Required for Full Swapping Station per day - KW 54			
Voltage - [V] 430			Output Power/day in KW			
Current - [I] 17			KW			
Power - [W] - KW/h 75			1800 Gen /day			
Power Availability 1.17			1772 use / day			
Fast Charging Station Load					ESS - Charging	
CBS	Voltage [V]	Power [kW]	Battery Capacity [kWh]	SOC Available	Required Current	Charging Time in Month charge /Cost
1. Tata Nano - EV	400	30.2	30.2	20	75.5	2.47 10 745.83
2. MG ZS EV	400	50	46.5	30	325	1.49 10 748.75
3. Tata Tigor EV	400	16.8	16	50	42	4.40 10 739.58
4. Mahindra eVerito	400	70	60	10	375	1.07 10 747.92
Total Power 167			Total Time 9		Total Revenue 2977	
Total Power 1670			KW power can charge / day		Total Time / %/ hr	Total Revenue 26794
Total Energy claculation						
Power sources	KW / day	KW / Month	KW / year	KW / 5 years		
Required full Power from Grid	1,817	54,510	653,203	3,316,025		
Solar	1,032	30,960	376,680	1,883,400		
Ess	576	17,280	210,240	1,051,200		
Grid	209	6,270	76,285	381,425		
Total Power saving / Day	1,608	48,240	586,920	2,934,600		
Total Generation Cost / Day 12rs / unit	21,804	654,120	7,958,460	39,792,300		
Total Generation Cost Saving / Day 12rs / unit	19,296	578,880	7,043,040	35,215,200		
Spending Amount	2,508	75,240	915,420	4,577,100		
Revenue						
Source	Per day	Per Month	Per Year	Per 5 Years		
Fast charging	26,794	803,813	9,779,719	48,898,594		
Battery Swapping	2,700	81,000	985,500	4,927,500		
Total Revenue	29,494	884,813	10,765,219	53,826,094		
Spending Amount	2,508	75,240	915,420	4,577,100		
Total Profit	26,986	809,573	9,849,799	49,248,994		

Conclusion

The proposed IoT-based architecture addresses critical challenges in EV charging infrastructure by enhancing efficiency, integrating renewable energy, and providing real-time monitoring.

This innovation paves the way for sustainable transportation and significant cost savings.